SPE-228886-MS

A Pragmatic Approach to Casing Deformation Remediation for Post Stimulation Wellbore Access Assurance Utilizing an Iterative Approach

Tim Manuel, Thruster Energy Corporation

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

Lateral wellbore sections below moderate to severely deformed regions are often abandoned as uncertainty of actual final recovery and variance associated with cost of intervention can ascend at a rapid rate. This is of particular importance when encountering instances of bedding plane slippage affecting the build section or liner top of the well; effectively restricting access to the entire wellbore and rendering the stimulated zones inaccessible. In this scenario, limited options for removal of temporary obstructions such as composite frac plugs are available and conventional bypassing options carry additional inherent operational risks when deployed.

The objective of this paper is to explore consistently successful methodologies for removal of casing impediments introduced during the process of multi-stage hydraulic fracturing (MSHF) in unconventional plays. This work intends to identify practical and repeatable methods for regaining access to sections of the wellbore below the deformed area with consistent results and a low variance of outcomes.

The solutions presented are summarized from a comprehensive review of operations performed on twenty wells wherein the nature of the deformations examined are principally tectonic factors such as natural fractures, fault slip and bedding-parallel slippage Type III (Casero & Rylance, 2020) in the Montney formation of the Western Canadian Sedimentary Basin (WCSB), though other deformation root-causes were present. However, this study focuses on remedial solutions post-event rather than exploring causation or event mitigation, as other works have, and this paper is thereby agnostic to root-cause factors present in the underlying sample cases.

Here we explore an iterative approach to remediation through which each deformation is measured and examined for severity then opened via a downhole machining process subject to the restraint of induced casing breach and additional contextual considerations are applied in selecting final methodologies for implementation. Access to the toe of the well is then able to be confirmed utilizing a risk-adjusted procedure selected using real-time assessment. Through evidence-based process refinements, the required steps are able to be quantified, normalized and optimized to a near 100% success rate.

The findings presented in this case history review serve as a demonstration that economic and measurable solutions to wellbore recovery following a MSHF induced deformation event can be obtained from the

perspective of service providers and well operators with the ultimate objective of reestablishing flow assurance from the toe of the well in the fewest steps possible.

Introduction

This work focuses on remediation of deformation caused most frequently by bedding plane slippage (BPS) and wellbore instability due to in-situ stress reacting with hydraulic fracturing stimulation resultant in casing integrity failure. This is mainly because the BPS depth is likely to affect casing close to the heel of the well, which is of significance because; a) the majority, if not all treated stages are isolated below the deformation, b) future tool deployments, interventions or pump/lifting equipment placement may become impacted, c) the deformation occurs late in the treatment schedule where the sunk cost of pumping the previous stages is large. All factors combined, wellbore recovery in this instance is imperative.

As shale gas and unconventional field development matures, these occurrences are only due to accelerate. A review of over 41,000 wells in Pennsylvania between 2000 and 2012 found a six fold higher incidence of casing and/or cement issues for shale gas wells than conventional wells (Ingraffea et al., 2014)

These play-specific instances are becoming more prevalent as infill activity increases in maturing fields, with large stress redistribution occurring when two producing parent wells are located in the same pay zone. This results in a shift in the magnitude of Shmin curve along target infill wells and orientation change of SHmax, further leading to unpredictability as field development continues (Sangnimnuan et al, 2019).

Even as instances of deformation become more common, workover planning in the damaged casing is not a new endeavour. As early as 1971, once damaged casing was encountered, three basic methods of proceeding remain relevant; (1) work through-tubing or to use undergauge tools; (2) use casing swages to restore the full diameter of the casing; or (3) to use long tapered milling tools to mill out the restriction, then isolate the damaged interval (McCauley, 1974). The expansion on remediation stems from modern technique refinement, equipment design and equipment material strength enabled during the US Shale Gas Revolution era expansion of North American E&P capital spending. This urgency to bring production online facilitated new technology due to the behemoth volume of wells completed and the appetite for operational efficiencies.

The work presented takes place almost exclusively in the Montney formation of the Western Canadian Sedimentary Basin (WCSB) and targets some regions particularly prone to BPS as a mode of failure. Casing is also observed to have been deformed through mild instances of ovality which increase the complexity of operations to remediate. These regions are governed by the Alberta Energy Regulator (AER) and BC Energy Regulator setting the requirements for casing integrity during pressure pumping and flowback operations.

The Decision to Address Casing Deformation

To perform a remediation on casing deformation, it must be economically viable given the marginal cost to achieve the marginal yield achieved from stages below. As such, the expected value of the resources can simply be expressed as a percentage of the well's NPV using terms such as EUR/lateral meter or EUR/proppant placed below the deformation depth. However the marginal cost curve for both time and equipment can be difficult to calculate for most well operators given a lack of any run history in these operations, let alone statistically significant data sets to produce reliable models. Very few operations personnel possess the relevant experience to direct the resources efficiently, further accelerating cost variance. Factored in also must be the inclusion of non-zero probability that an unintended casing exit may occur, which then becomes difficult or nearly impossible to recover from.

The operation also must take into consideration future planning of the field and the impact on casing integrity derived from machining material off the ID of the casing. While usage of the API 5C3 model for calculating performance of casing tubulars is standard, the traditional API method has been demonstrated to understate the collapse strength after the modernization of casing manufacturing whereby the adjusted factors of P-110 and Q-125 casing can be shifted utilizing the Ultimate Limit Strength (ULS) method which more closely resembles observed collapse values (Brechan et al., 2020). This method results in an acceptable increase to tubular D/t ratios in a given casing string while maintaining post treatment integrity without increasing the likelihood of plastic collapse under normal production operation conditions. This remains a consideration for future field development and may require further mitigation and monitoring as child well stimulation occurs.

Risk mitigation is key, and while following best practices during pressure pumping operations can often alleviate unintended consequences from stress migration, completely avoiding the deformation may not be feasible. Most notably when the reduction in casing ID exceeds 10% of nominal ID, the likelihood of adjusting completions and millout strategies reaches a certainty.

A modern additional solution involves avoidance through (relatively) new technology; the operator may choose to utilize dissolvable frac isolation plugs from treatment onset:

Dissolvable plugs can yield mixed results as the typical failure rate (read: plug slips, soft sets, bypass) on dissolvable plugs can erode the value of the treated interval isolated by dissolvable means. Individual operators may categorize plug failures differently, but it is not uncommon for failure rates of dissolvable plugs to be in excess of 30%. Even in highly salinated wellbore fluids, the ion interaction required to dissolve the plug may not occur in the absence of flow. Without chemical tracers, this leaves the well operator little means of assurance that flowback is occurring from stages below deformation as indicators can be difficult to identify in the early stages of flowback and producing of the well. A high reliance is placed on the dissolution process taking place to leave an unobstructed flow path, though direct observation of this is unattainable.

This method does, however, fail to address the future ability for intervention work past that deformed depth and the ability to run tools like ESPs below the deformed depth.

Historical Solution: Tapered Milling

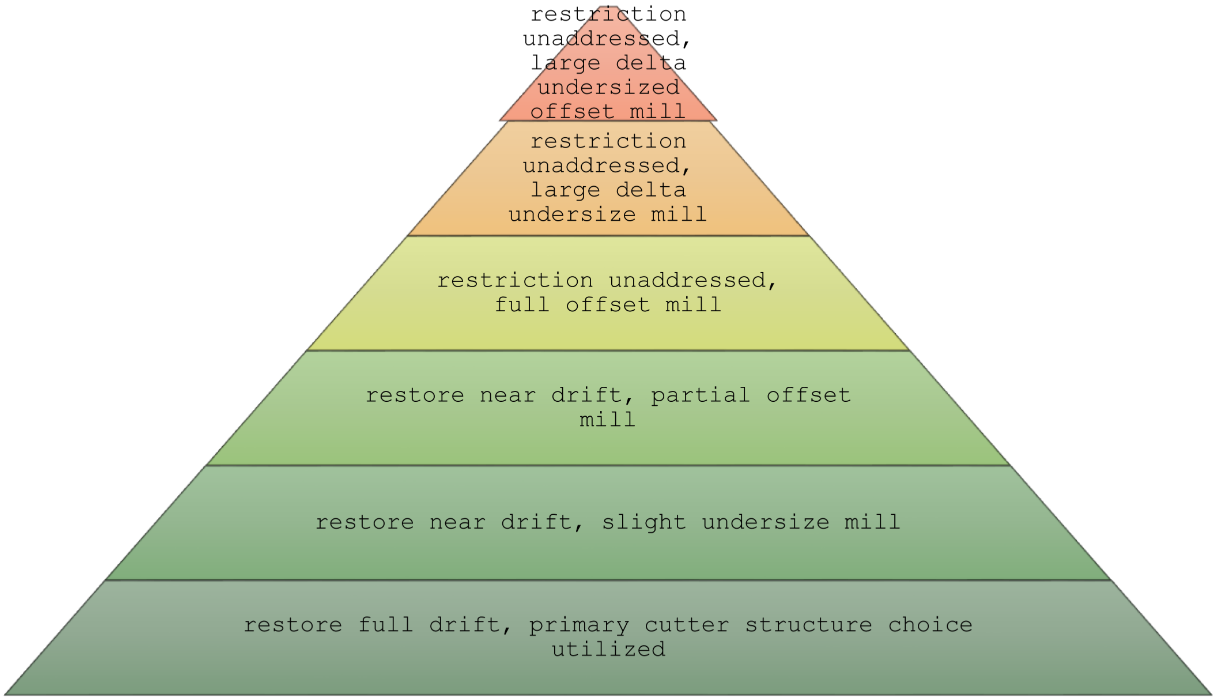
As previously described, taper milling was historically a method of remediation. A tapered mill features a continual change in OD from the tip to the shank, where the slope can be customized to suit varying applications. While taper mills have suitable applications, addressing casing deformation with them has proven to be cumbersome as the taper mill will bind into the casing and stall the mud motor before performing tangible work. The reason for this is too great of mill surface engagement is likely, therefore controlling the mill to casing interface is extraordinarily difficult and small movements of the CT string result in unpredictable torque applications, resulting in touch-stalling of the motor with very little injector movement. This result is typified by cycled out CT fatigue and ultimately an unsuccessful mill run where the coil must be tripped to surface and slip/cut to move the high cycle point off the deformation depth. This process is repeated with few ways to verify progress as CT runs take place.

Risk Matrix

Typical risk profile of running downhole assemblies into any completed well are amplified by casing deformation as any reduction in the casing ID will constrict the deployment of fishing assemblies in the event of breakage, controlled release of stuck tools or other form of unexpected tool loss that may be caused by the change in geometry.

Risks during remediation include getting the BHA stuck, exit casing, breaking tools/leaving an unrecoverable fish. These risks are again amplified in absence of caliper logging data. The design of the BHA must take into consideration the length, severity and orientation of the deformation so the cutting structures can be localized and applied to the surface of the deformation without causing deflection in the assembly. This is a critical piece of information to ensure the BHA remains centralized and does not inadvertently sidetrack during extended hours of casing milling.

Risk Stack: Casing Milling Outcome with Highest:Lowest Probability of Unintended Milling Outcome



These risks may be amplified given the number of stages, type of frac plugs and anticipated hours spent with tools below deformed area. Approaching this blind (absent casing caliper logs) will also increase the operational risk profile.

Table 1—Properties of Various Applicable Casing Types

Casing Size (mm, in)	ID (mm,in)	Steel Grade / Type	API Wall thickness (mm,in)
139.7mm; 34.23kg/m	118.6mm, 4.669″	SM95S	10.54mm, 0.415″
127mm; 35.86kg/m	98.4mm, 3.874″	P-110 CYE	7.92mm, 0.312″
127mm; 35.86kg/m	98.4mm, 3.874″	P-110 SML IC	7.92mm, 0.312″
114.3mm; 22.47kg/m	97.2mm, 3.826″	VA-EP-P110	8.56mm, 0.337″
114.3mm; 22.47kg/m	97.2mm, 3.826″	P-110 SML EC	8.56mm, 0.337″
114.3mm; 22.47kg/m	97.2mm, 3.826″	P-110 SML EC	8.56mm, 0.337″

Equipment

Step mills are specially designed to machine out small, controlled areas of the deformed casing at a predictable rate. Key components of the step mill are the guide, transitions and stages. The stages will be of predetermined length to separate the transition areas and ensure multiple transitions do not engage the casing simultaneously, rapidly increasing the torque load on the motor and inducing a stall. Stalls when step milling occur quickly. The motor is brought into the deformation free-spinning with no torque load. This brings the mill to high rpms and as a large mass, creates high inertia prior to engagement. Several key components within the motor assembly must be capable of handling the extreme torsional load of this type of stalling combined with heavy overpulls to free the stuck mill.

Figure 1—example of step mill

Mud motors must be capable of high revolutions, in excess of 500 RPM, as the step milling process requires a lathe-like interaction between the mill and the casing which will be impeded by a lack of rotation.

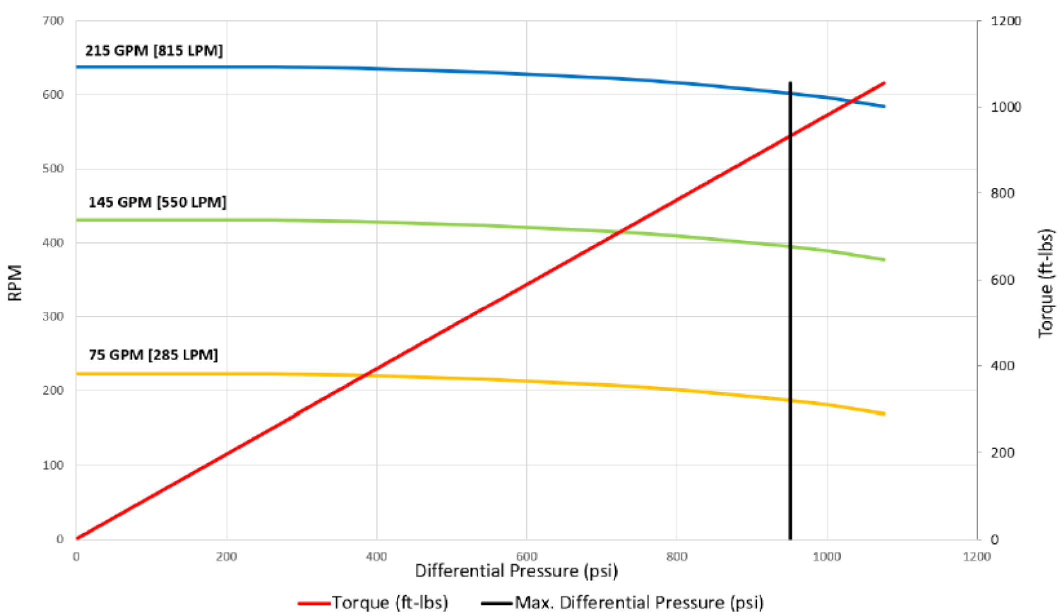


Figure 2—performance curve of suitable motor assembly

The most favourable conveyance method is coiled tubing due to the continuous circulation and fine control of the injector. The addition of surface rotary equipment can overkill and complicate the operation, leading to unnecessary variability where static conditions are appropriate. Surface equipment must be suitable to permit high flow rates without efficiency losses. It is recommended to utilize high grade CT strings at least 2.375″ in diameter that are capable of high pressure cycling and have ability to accept high axial loads in the event of stalling if the BHA becomes bit stuck with the potential for large surface string weight. The injector must be suitable (>130K recommended) for these large maximum pulls and able to provide snub force on the CT string to transmit adequate WOB. In addition to strength, the injector's fine speed controls must be configured for small, consistent adjustments to pipe depth to allow precise control of WOB applied during the operation.

Fluid pumping equipment must possess sufficient kilowatt power to provide high pumping rate at potentially high pressure due to charged nature of wellbore immediately post-stimulation. Upwards of 1,000hp x2 twin pumping units are preferred for this type of operation.

Cuttings will be fine metal shavings – pigtails with very little thickness. These cuttings are easily carried to surface so while annular velocities remain a consideration, they become secondary as the pump rate required to achieve sufficient RPM will have a secondary effect of maintaining high AV during step milling operations.

Surface flowback equipment should possess sufficient filtering equipment to capture this debris for analysis throughout the operation, checked at regular intervals to evaluate progress and ensure no warning signs become present.

As evident by the equipment requirements, this process is one of precision and as such the use of extended reach agitation tools should be avoided when possible. Eccentric motion or axial pulsations inevitably further complicate an already delicate operation.

Use of Offset or 'Swing' Mills

An offset mill is an important component of dealing with casing deformation, but it is not an absolute silver bullet for all applications. Its use must be carefully evaluated for suitable operations. By offsetting the mill head from the center of the shank or fish neck, a swing path is created, allowing the mill to address the full surface area of plugs below the deformation even with a reduction is the size of the head. While the physical contact is increased when compared to under sizing the mill, considerations such as reduced weight on bit must be observed to avoid stalling the motor while planting the mill into the plug/obstruction. Although presenting additional challenges, this is still favourable to running a simple undersized mill, which is much more likely to drill through the center, or "core" the plug which will also lead to planting the BHA in the well.

Two stages of offset mills are commonly utilized, partial and full offset. Partial offset mills are a good means of dealing with reduced casing ID. If an offset mill is required, this is the preferred configuration as a full offset is much more aggressive, prone to stalling and unintended BHA damage. In contrast, the geometry of the partial offset mill increases the number of impact points on the casing wall, which means it will sweep the casing in more controlled manner. Closer to center, the eccentric forces applied to the motor's drive shaft are less severe than utilizing a full offset mill. For a more broad swing radius a full offset mill may be required. The full offset mill features one fully flat surface running the length from the tip of the carbide to the pin connection. This design generates the greatest eccentric forces due to the cam-over design. This configuration should be limited to low plug counts and short runs as they rapidly fatigue the motors' bearings and drive assembly.

Procedures

Diagnostics: a diagnostic run is critical to properly addressing the deformed section with the optimal number of step milling runs. This data will make known the exact severity, orientation and length of each deformed section. A well cleaning run is highly recommended prior to any diagnostic run to eliminate loose debris which can impede accurate measurement from the caliper arms. It is also recommended during this run to place a metal-metal friction reduction chemical throughout the lateral section as caliper runs will necessarily be completed without extended reach tools to mitigate friction challenges. Coil friction can result in stick and slip problems which can degrade the quality of the diagnostics while recording. Step milling involves specially designed milling bit featuring steps and transitions downhole metal lathe removing a very small amount of material with each pass, therefore the accuracy of this data is extremely important.

Framework for Developing Procedures

No Two Deformations are the Same: Identification and Root Cause

Understanding the nature of the deformation will drive probabilistic reasoning to solve problems such as; is this event isolated? Where else might there be casing issues? Can the severity of the event driven deformity be correlated to the magnitude of the event (or by some factor thereof)?

Assessment & Measurement

Did the event occur during treatment of an off azimuth well? Was the anomaly detected due to previously unseen stick-slip of wireline equipment during pump down, or were spent wireline tools causing overpulls after preparing a zone? How many intervals have been stimulated since indicators were observed? Answering these questions will help identify the target area to be measured and recorded with caliper logging tools.

Log Evaluation & Scenario Planning

What is below the area of interest? What is the ideal method of accessing stages below this part of the wellbore and what is my risk-adjusted minimal accepted method? How long will I spend under various scenarios before returning tools to surface to evaluate the run results? Programming these criteria will eliminate noise in decision making and isolate controllable variables.

Execution and Real Time Adjustments

A relatively clean well path must be established for a step mill to successfully machine the casing as intended. Additional clean up runs might be pertinent to ensure large pieces of debris do not inhibit the ability of the tools to perform as expected and will likely net faster results than attempting to work around obstacles that will ultimately result in BHA sticking. Real time adjustments will be made when a BHA run does not behave as expected. Evaluation of the mill wear is critical when addressing deviations from the original procedure.

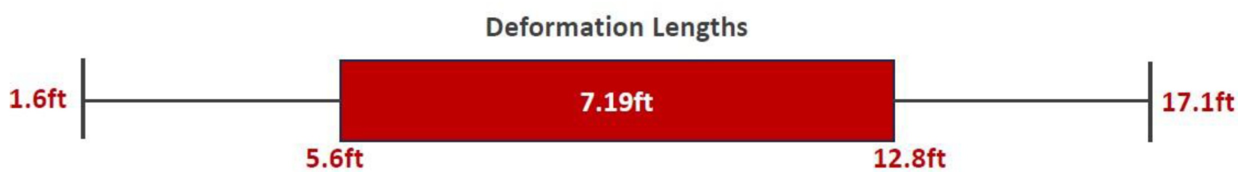


Figure 3—assessment of deformation lengths; minimum, maximum, average and 1σ

Table 2—Key Metrics from Wells with Addressed Casing Deformation

Well	Diagnostic Run	Step Milling Runs	Number of Deformations	Composite Plugs Below Deformation	Total Stage Count	Drift Restored	TD Achieved?	Operation Hours
1	Yes	1	1	16	38	Yes	Yes	178
2	No	7	1	3	27	No	Yes	214
3	No	3	1	23	28	Yes	Yes	96
4	Yes	6	2	18	31	Yes	Yes	185
5	Yes	3	1	39	47	Yes	Yes	126
6	No	3	2	37	41	Yes	Yes	68
7	Yes	13	4	23	25	No	Yes	314
8	Yes	3	1	18	20	Yes	Yes	92
9	Yes	9	3	41	46	No	Yes	299
10	Yes	3	1	16	16	Yes	Yes	73
11	No	2	1	17	25	Yes	Yes	69
12	Yes	5	1	26	29	No	Yes	84
13	Yes	3	1	27	31	Yes	Yes	84
14	Yes	3	1	14	37	No	Yes	131
15	No	2	1	19	40	No	Yes	93
16	No	9	3	32	45	Yes	Yes	212
17	No	3	1	32	45	Yes	Yes	99
18	No	7	5	22	25	No	Yes	171
19	No	2	2	13	32	No	Yes	88

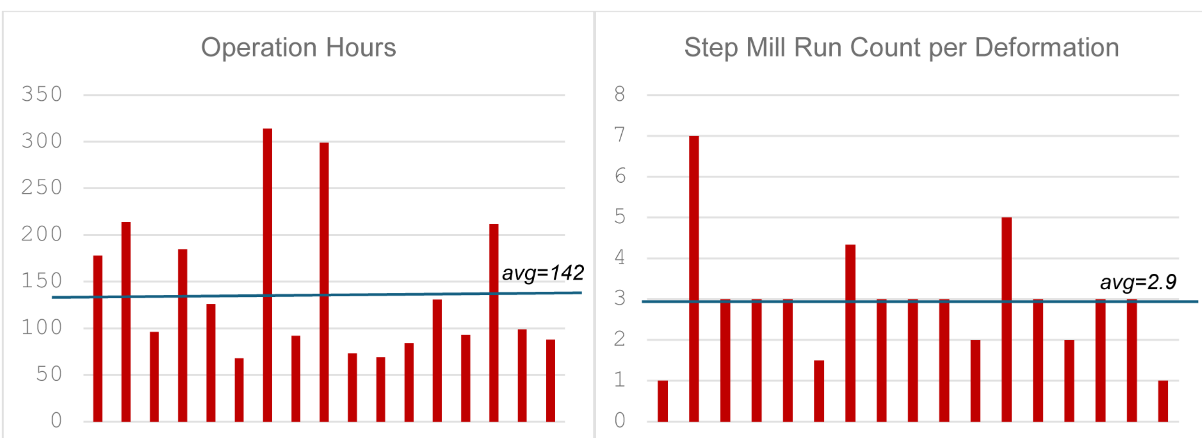


Figure 4, 5—plot of operational hours and run count, respectively, with sample averages

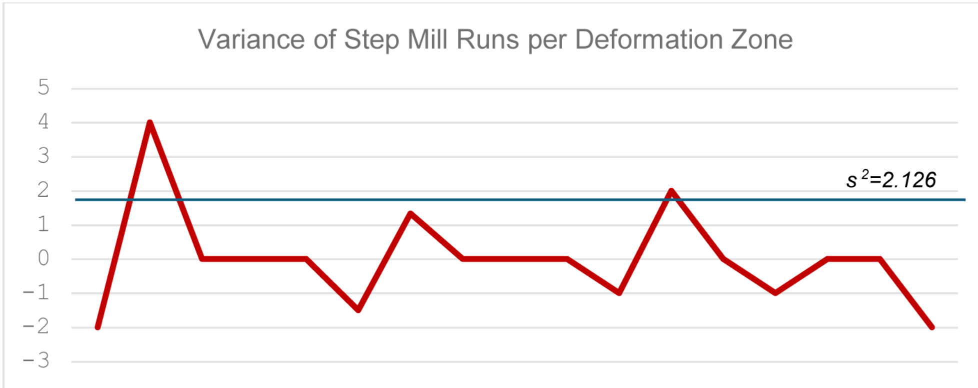


Figure 6—expected value of run variance for step milling operations, per deformation zone

Case Analysis

Case 1 – Simple Remediation

Well 1 is a horizontal wellbore drilled to a measured depth of 5,103m and a vertical depth of 1,946m. The well was completed with 26 composite bridge plugs used for stage isolation. Casing utilized was 4.5″ VA-XP-SS95 to a depth of 2,022m with the remaining 3,081m of lateral wellbore cased with 4.5″ VA-EP-110. The deformation occurred 156m into the lateral section of the well at a depth of 2,386m. The nominal casing ID was 97.2mm.

Upon detection of the deformation event, a 24-arm caliper logging tool was deployed from 2,607m – 2,269m for a total of 338m of evaluated wellbore. The deformation was measured from 2,386.8m – 2,389.0m for a total of 2.2m of deformed length. The maximum cross-well diameter restriction was measured at 82.56mm, a 15% reduction in ID.

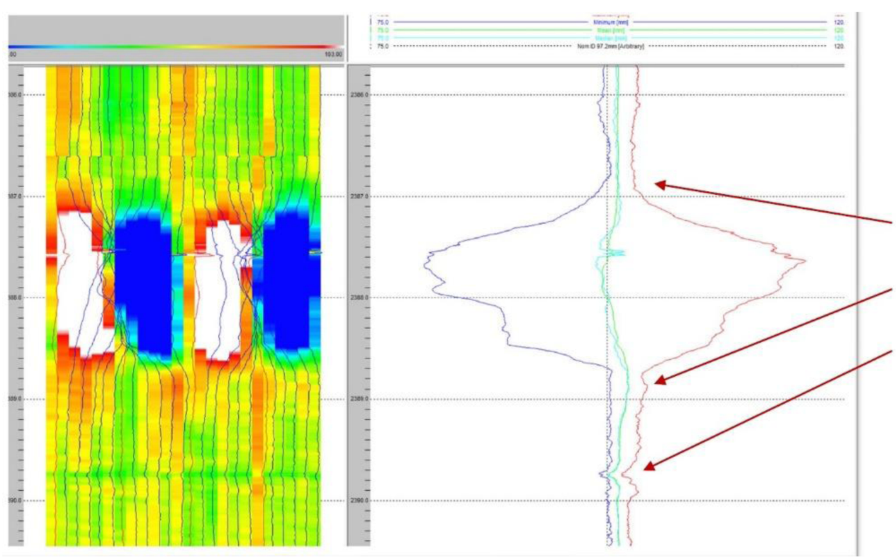


Figure 7—caliper log results from case 1 deformed section

An initial effort to drill out all composite bridge plugs, including those below the deformed section, was completed with an offset mill measuring 85x97mm, where the 85mm face of the mill should have entered the deformed section. The offset mill successfully drilled the first two plugs but was unable to pass through the restriction. Indications of this are shown in figure ___, depicted the worn shoulder of the mill on the first pass.

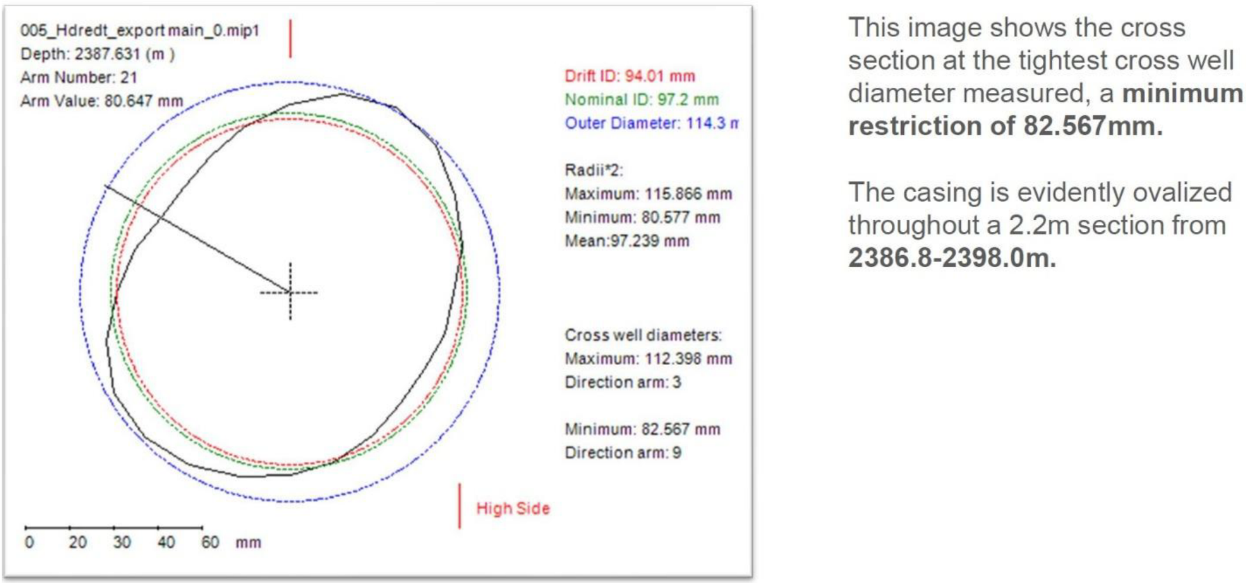


Figure 8—cross section of deformation

Figure 9—post-run image of offset mill face with indications of wearing on shoulder.

A total of 23 minutes was spent attempting to orient the mill to pass through the deformation before the BHA was pulled to surface and inspected. Three step milling runs took place with sequentially larger step mills where 5.8 hours, 7.8 hours, and 1.6 hours were spent milling with differential torque engagement on the deformation respectively. Upon completion of the third run, it was confirmed that the ID had been restored to 92mm or within 5% of nominal casing ID (2% reduced from drift casing ID). This allowed the final plug milling run to be conducted with a 92mm tricone bit ensuring a smooth milling operation and small, manageable debris throughout the operation.

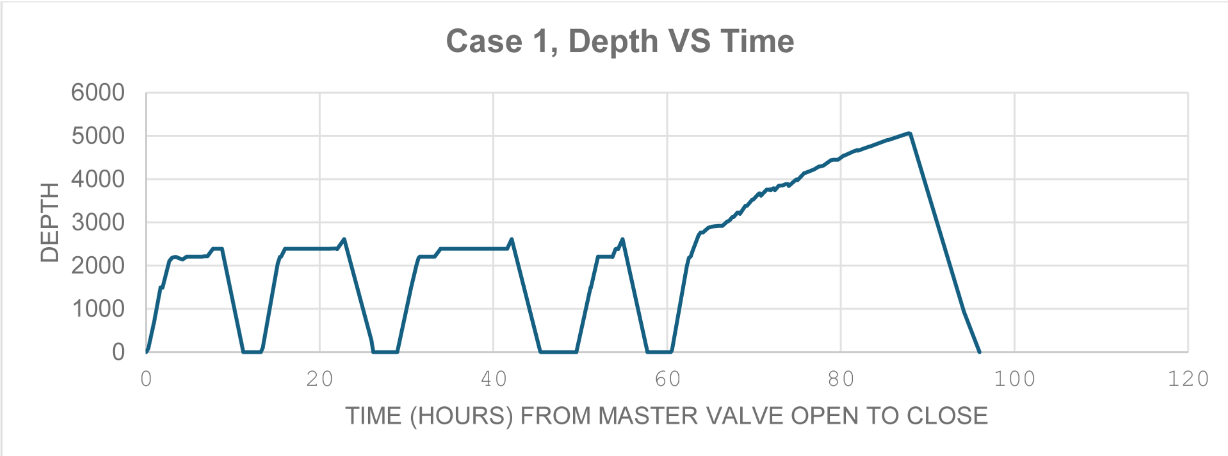


Figure 10—chart of case 1 operating time vs depth, by run

Table 3—case 1 run summary

	Run	Run Depth (m)	OD Size	Total Hours
1	Offset Milling BHA	2,389	85mm- 97mm	11.2
2	Step Milling	2,610	76mm – 85mm	13
3	Step Milling	2,610	73mm – 92mm	16.5
4	Step Milling	2,610	73mm – 92mm	8.2
5	Tricone Milling BHA	5,060	92mm (Varel)	35.5
			TOTAL	84.4

This operation took place over 6 days when considering trip time and surface preparation time. The result is highly successful as the desired outcome is obtained with an optimized count of runs over a short duration. Results are quickly attainable when knowledge of casing anomalies is high and a large degree of certainty can be assigned to run selection and decision making. Accurate downhole visualisation means the steps required can be measured and deployed succinctly and each planned run can be prepared following the success of the previous run. This reduces delays and downtime between runs as surface activities & maintenance are kept minimal and next-run decision making has already considered most likely outcomes from the current run.

Case 2 – Complex Multiple Deformations, No Diagnostic Run

Well 2 is a horizontal wellbore drilled to a measured depth of 3,811m and a vertical depth of 1,966m. The well was completed with 18 composite bridge plugs used for stage isolation. Casing utilized was 4.5″ VA-SS-95 to a depth of 1,947m with the remaining 1,864m of lateral wellbore cased with 4.5″ VA-EP-110. Multiple deformations occurred in the lateral section of the well at depths of 2,453m, 2,470m, 2,502m, 2,513m and 2,750m. The nominal casing ID was 97.2mm.

The first run into this well was with an offset mill swinging the nominal casing ID. Composite plugs down to stage 5 were successfully removed with this BHA to a depth of 2,384m. Many of the initial stages had been drilled out on a previous run, so the 6 remaining plugs above the deformation were a contributing factor in selecting the offset mill. The underlying reasoning that the well could potentially be brought fully online in a single trip if the 85mm face were able to pass through the deformed section to remove the last 3 plugs. This risk was further mitigated by the low plug count below the deformation and the short nature of the well limiting exposure of the BHA to eccentric impacts throughout the duration of the run. However the BHA tagged at 2,453m identifying the depth of concern. Unable to pass this depth, the BHA was removed from the well for evaluation. Post run 1, the mill was determined to be 2mm under gauge so a 73-92mm step mill was installed to immediately address the area. The large change across many steps was designed as a blanket approach when the deformed section has not been previously measured. This run saw small but insignificant intervals of work while running in to the known anomaly. This can be attributed possibly to casing ovality, additional deformation, or simply plug debris and sand remaining in the well from the previous run. At 2,410m a motor stall was induced while tripping CT in-hole and this section of casing was milled for 2 total hours making 1m of progress before it was decided to pull to surface and inspect tools. On surface it was observed that wear existed at all mill transitions and the final section gauged 92mm, therefore the subsequent run was decided to be with a smaller and more concise size of 76-85mm.

This second step mill run did not see the restriction at 2,410m but did tag at 2,453m which drilled through 2.15m in 5.7 hours. This mill continued in the well and saw a small amount of work at 2,470m over a 1.7m length. This process continued at 2,502m where it was no longer able to make new depth, stalling very quickly after engagement so the BHA was again pulled to surface for inspection.

Utilizing this recursive procedure, the operator is able to evaluate wear and determine run n+1 mill sizing on-the-fly in the absence of credible logging data.

Ultimately, 9 runs were performed in total over 7 days, removing the bottom 3 plugs and accessing the first 4 toe zones in the well. It is difficult to determine if efficiencies in sizing selection may have eliminated a run or possibly more, given the unknown nature of the deformation from the onset of the operation.

This method is also resource intensive as key decision makers must be made available and reach collective agreements as each run is examined for evidence of downhole progress.

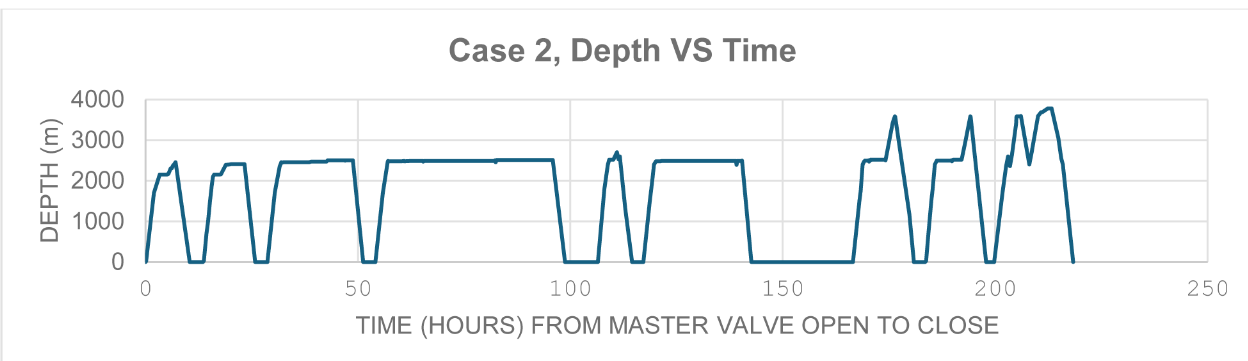


Figure 11—chart of case 2 operating time vs depth, by run

Table 4—case 2 run summary

	Run	Run Depth (m)	ODSize	Total Hours
1	Offset Milling BHA	2,453	85mm- 97mm	10.4
2	Step Milling	2,410	73mm – 92mm	12.3
3	Step Milling	2,503	76mm – 85mm	22.5
4	Step Milling	2,514.6	68mm – 80mm	44.6
5	Step Milling	2,570	68mm – 78mm	8.1
6	Step Milling	2,491	76mm – 85mm	25.4
7	Step Milling	3,582	76mm – 83mm	14.4
8	Step Milling	3,582	76mm – 85mm	14.3
9	Offset Milling BHA	3,787	83mm – 93mm	18.6
			TOTAL	170.6

Conclusion

Awareness surrounding principle drivers of fracture induced casing deformities is rapidly increasing, however this issue will persist in unconventional well development. A strong foundation of standards in addressing casing deformation when it occurs can lead to remediation in a predictable manner if systematically followed whereby a high degree of precision is required to obtain consistent results. Each unique scenario must be evaluated and planned utilizing framework that factors a multitude of variables that are outlined here in a highly endogenous system.

Key learnings include the significance of post run step mill evaluation and adequate reserve equipment to deploy given evolving downhole conditions. Proper wholistic risk assessment weighted by all contributing factors including future planned operations. The wellbore can be recovered with high IRR on remaining stages given the scope of the operation. This is best addressed with downhole modeling of the deformed area with reliable caliper data as a solid foundation for decision making.

Terminology & Nomenclature

API	American Petroleum Institute
Az	Azimuth
BHA	Bottom Hole Assembly
BPS	Bedding-Plane Slippage
EUR	Estimated Ultimate Recovery
FN	Fishneck
ID	Inner Diameter
IRR	Internal Rate of Return
NPV	Net Present Value
OD	Outer Diameter
RPM	Revolutions per minute
Shmin	Minimum horizontal stress
SHmax	Maximum horizontal stress
TD	Total Depth
ULS	Ultimate Limit Strength
WCSB	Western Canadian Sedimentary Basin
WOB	Weight on Bit
WT	Wall Thickness

References

Almulhim, Bassam Khalid, Ashby, Scott, Alobaid, Hashem, and Abdulmuin Alyousef. "Evaluation and Study of Dissolvable Frac Plugs on Horizontal Well Fracturing, Plug and Perf Completions." Paper presented at the SPE Annual Technical Conference and Exhibition, San Antonio, Texas, USA, October 2023. doi: https://doi.org/10.2118/214841-MS

Brechan, Bjorn, Kornberg, Even, Sangesland, Sigbjorn, and Stein Inge Dale. "Well Integrity Model - Klever & Tamano Collapse." Paper presented at the SPE/IADC Middle East Drilling Technology Conference and Exhibition, Abu Dhabi, UAE, January 2018. doi: https://doi.org/10.2118/189395-MS

Han, G., Arshad, W., and J. Diessl. "Casing Damage in Hydraulic Fracturing: Geomechanical Perspectives." Paper presented at the 55th U.S. Rock Mechanics/Geomechanics Symposium, Virtual, June 2021.

Ingraffea, A.R., Wells, M.T., Santoro, R.L., and S.B. Shonkoff, 2014. Assessment and risk analysis of casing and cement impairment in oil and gas wells in Pennsylvania, 2000–2012. In *Proceedings of the National Academy of Sciences of the United States of America, 29 July 2014*. **11**(30): 10955–10960. https://doi.org/10.1073/pnas.1323422111

McCauley, T.V.. "Planning Workovers in Wells With Fault-Damaged Casing-South Pass Block 27 Field." *J Pet Technol* **26** (1974): 739–745. doi: https://doi.org/10.2118/4607-PA

Rylance, M., and J. M. Montoza. "A Systematic Approach for Processing, Prioritising and Progressing Casing Deformation Understanding in Unconventional Reservoirs." Paper presented at the SPE Hydraulic Fracturing Technology Conference and Exhibition, The Woodlands, Texas, USA, February 2025. doi: https://doi.org/10.2118/223529-MS

Sangnimnuan, Anusarn, Li, Jiawei, Wu, Kan, and Stephen Holditch. "Impact of Parent Well Depletion on Stress Changes and Infill Well Completion in Multiple Layers in Permian Basin." Paper presented at the SPE/AAPG/SEG Unconventional Resources Technology Conference, Denver, Colorado, USA, July 2019. doi: https://doi.org/10.15530/urtec-2019-972

Uribe-Patino, J. A., Casero, A., Dall'Acqua, D., Davis, E., King, G. E., Singh, H., Rylance, M., Chalaturnyk, R., and G. Zambrano-Narvaez. "A Comprehensive Review of Casing Deformation During Multi-Stage Hydraulic Fracturing in Unconventional Plays: Characterization, Diagnosis, Controlling Factors, Mitigation and Recovery Strategies." Paper presented at the SPE Hydraulic Fracturing Technology Conference and Exhibition, The Woodlands, Texas, USA, February 2024. doi: https://doi.org/10.2118/217822-MS

Zhou, Jing, Huang, Hai, Deo, Milind, and Shu Jiang. "A New Physics-Based Modeling of Multiple Non-Planar Hydraulic Fractures Propagation." Paper presented at the SPE/AAPG/SEG Unconventional Resources Technology Conference, San Antonio, Texas, USA, July 2015. doi: https://doi.org/10.15530/URTEC-2015-2170875